WEATHER APP – PROJECT REPORT

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Course: Introduction to Problem Solving and Programming
Platform: VITyarthi – Build Your Own Project

1. Introduction

Weather conditions play an important role in day-to-day activities such as travelling, outdoor planning, and scheduling events.

The **Weather App** is a simple Python based desktop application developed to provide current weather information for a city entered by the user.

The application uses a graphical user interface created using Tkinter. It first converts the entered city name into latitude and longitude using the Open-Meteo Geocoding API. These coordinates are then sent to the Open-Meteo Weather API to retrieve current weather information.

The application displays useful information such as:

- Temperature
- Humidity
- Current weather condition

The project demonstrates the use of Python programming concepts, GUI development, API integration, JSON data handling, functions, and exception handling.

2. Problem Statement

Problem

Users often need quick weather information for travel and daily planning. Searching through multiple websites can be inconvenient.

Solution

A simple Python application where the user enters a city name and gets current weather information instantly.

3. Functional Requirements

The project has three main functions:

1. **City Search** – User enters a city name.
2. **Weather Retrieval** – APIs retrieve location and weather data.
3. **Result Display** – Temperature, humidity, and weather condition are displayed.

The reference report similarly organizes the project around three main functional modules.

4. Non-Functional Requirements

- **Performance:** Weather should load within a few seconds.
- **Reliability:** The program should not crash during API errors.
- **Usability:** The interface should be simple and easy to use.
- **Error Handling:** Invalid cities and network errors should be handled properly.

These match the same categories used in the reference report: performance, reliability, usability, and error handling.

5. System Architecture

User
↓
Tkinter GUI
↓
City Name
↓
Open-Meteo Geocoding API
↓
Latitude & Longitude
↓
Open-Meteo Weather API
↓
Weather Data
↓
Display Result

6. Design Diagrams

6.1 Use Case

User
1. Enter City
2.Click Get Weather
3.View Weather Result

6.2 Workflow

Start
↓
Enter City
↓
Get Coordinates
↓
Fetch Weather
↓
Display Result
↓

End

6.3 Sequence

User → GUI → Open-Meteo Geocoding API → Open-Meteo Weather API → GUI → User

7. Design Decisions

- **Python:** Simple and easy to use.
- **Tkinter:** Used for the desktop GUI.
- **Requests:** Used for API communication.
- **Open-Meteo Geocoding API:** Converts city names into latitude and longitude.
- **Open-Meteo Weather API:** Provides current weather information without requiring an API key.

8. Implementation

The application is contained in one main file:

Weather.py

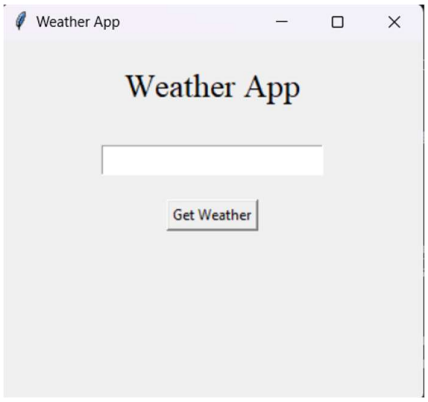
Libraries used:

import tkinter as tk

import requests

9. Screenshots

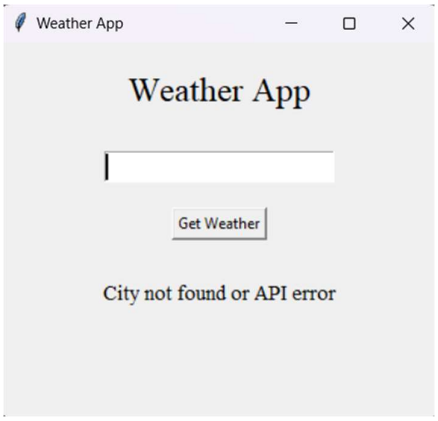
1.Entering the city name



2.weather condition of city



3.Error generated when no city is entered and pressed get weather



10. Testing Approach

- **Standard Test**

Enter "Bhopal" and click **Get Weather**.

Expected: Temperature, humidity, and weather condition should appear.

- **Invalid City Test**

Enter an invalid city such as:

abcdxyz123

Expected: Error message should appear.

- **Network Test**

Disconnect the internet and search for a city.

Expected: The application should handle the error without crashing.

11. Challenges Faced

• Understanding API integration.

• Converting city names to latitude and longitude.

• Understanding and processing JSON responses.

• Handling invalid city names and network errors.

12. Learnings

Through this project, I learned:

- Tkinter GUI development
- API integration
- JSON data handling
- HTTP requests
- Functions

- Exception handling

13. Future Enhancements

- 5-day forecast
- Weather icons
- Search history
- Automatic location detection
- Celsius/Fahrenheit option
- Improved GUI

14. Conclusion

The Weather App is a simple Python desktop project that allows users to check current weather for any city.

It demonstrates important Python concepts such as **Tkinter, functions, API integration, JSON processing, and exception handling**.

15. References

1. Python Documentation

2. Tkinter Documentation

3. Requests Documentation

4. Open-Meteo Weather API Documentation

5. Open-Meteo Geocoding API Documentation